

Exercise Sheet 2  
Lecture “Cybersecurity”  
Summer Term 2025  
Practical Exercise Symmetric Cryptography

## Vernam Cipher (One-time pad and Stream Cipher)

The Vernam cipher is an extension of the Vigenère cipher, where the key used for encryption is the same length as the plaintext. Such a method is considered insecure by modern standards. However, if the key is generated using a cryptographically strong random number generator, Vernam becomes a stream cipher. The security then depends on the randomness of the generator. If the key is truly random, the method is referred to as a one-time pad.

|   |   |   |   |   |    |   |    |   |    |      |
|---|---|---|---|---|----|---|----|---|----|------|
| A | 0 | F | 5 | K | 10 | P | 15 | U | 20 |      |
| B | 1 | G | 6 | L | 11 | Q | 16 | V | 21 |      |
| C | 2 | H | 7 | M | 12 | R | 17 | W | 22 | Z 25 |
| D | 3 | I | 8 | N | 13 | S | 18 | X | 23 |      |
| E | 4 | J | 9 | O | 14 | T | 19 | Y | 24 |      |

Table 1: Caesar Alphabet, assigned to numbers (0-25)

Encryption:

1. Each letter in the plaintext is assigned a number from 0 to 25 (see Table 1).
2. The corresponding key value is added to each letter (plaintext + key).
3. Subtract 26 if the result is greater than 25. Otherwise, leave it unchanged.

Decryption is the inverse process.

Formal Definition:

$Z_{26}$  denotes the set of numbers from 0 to 25, with a unique letter assignment as described in Table 1.  
Let  $k \in Z_{26}$  be the key.  
Encryption: For each letter  $x$  in the plaintext, apply the function:  
 $e(x) = (x + k) \bmod 26$   
Decryption: For each letter  $y$  in the ciphertext, apply the function:  
 $d(y) = (y - k) \bmod 26$

Example:

|    |    |    |    |    |    |   |    |    |    |    |
|----|----|----|----|----|----|---|----|----|----|----|
| C  | Y  | B  | E  | R  | S  | E | C  | U  | R  | E  |
| 2  | 24 | 1  | 4  | 17 | 18 | 4 | 2  | 20 | 17 | 4  |
| K  | E  | Y  | R  | A  | N  | D | O  | M  | K  | I  |
| 10 | 4  | 24 | 17 | 0  | 13 | 3 | 14 | 12 | 10 | 8  |
| M  | C  | Z  | V  | R  | F  | I | Q  | G  | B  | M  |
| 12 | 2  | 25 | 21 | 17 | 5  | 8 | 16 | 6  | 1  | 12 |

### Exercise 1.

1. Encrypt the message: **VORLESUNG** using the Vernam cipher with the key: **SECUNIDUE**
2. The method shown is a (weak) variant of the one-time pad. Give the number of possible keys for a Vernam cipher with plaintext length **5**. How many keys exist for a plaintext of length **n**?
3. Assume a CPU with 2 cores and 4 GHz is fully dedicated to key computation for a message of length **1024**. Each key computation requires 4 operations. 1 GHz equals 1,000,000 operations per second. How long does it take for an attacker to brute-force a key with this CPU?
4. *Multiple Encryption*: A popular method to increase security is to apply the encryption multiple times. For example, the **DES** cipher can be made secure by applying it three times.  
A double encryption can be expressed as:  $y \equiv e_{k_2}(e_{k_1}(x))$   
How does double encryption affect the key space  $K$  in the Vernam cipher? Provide a general formula based on message length **n**.

## Malleability of Stream Ciphers

Malleability describes a property of a cryptosystem where changes to the ciphertext (y) can result in controlled changes to the plaintext (x) without knowing the plaintext. The attacker can modify the ciphertext and achieve predictable changes in the decrypted plaintext.

### Exercise 2.

1. Explain why stream ciphers are inherently malleable and how an attacker can exploit this to make targeted changes to the plaintext.
2. Suppose you are an attacker and have hacked into a Bitcoin broker's system overnight. The broker transfers 1 Bitcoin to you (currently worth 40,000 EUR). The number of Bitcoins is stored as a 16-bit integer (in C: **short**). How can you modify the value to become a millionaire?
3. Suppose you are a malicious admin of a streaming service with free employee access. You want to watch all premium content without paying. So, you change the ASCII value of 'm' (employee) to 'p' (premium). The database entry is encrypted as a ciphertext  $y('m')$  (stream cipher) with binary value 100 0000. What ciphertext (binary) must you set to view premium content? (Hint: known plaintext attack)

# 1 ASCII Table

| Decimal | Hex | Char                   | Decimal | Hex | Char    | Decimal | Hex | Char | Decimal | Hex | Char  |
|---------|-----|------------------------|---------|-----|---------|---------|-----|------|---------|-----|-------|
| 0       | 0   | [NULL]                 | 32      | 20  | [SPACE] | 64      | 40  | @    | 96      | 60  | `     |
| 1       | 1   | [START OF HEADING]     | 33      | 21  | !       | 65      | 41  | A    | 97      | 61  | a     |
| 2       | 2   | [START OF TEXT]        | 34      | 22  | "       | 66      | 42  | B    | 98      | 62  | b     |
| 3       | 3   | [END OF TEXT]          | 35      | 23  | #       | 67      | 43  | C    | 99      | 63  | c     |
| 4       | 4   | [END OF TRANSMISSION]  | 36      | 24  | \$      | 68      | 44  | D    | 100     | 64  | d     |
| 5       | 5   | [ENQUIRY]              | 37      | 25  | %       | 69      | 45  | E    | 101     | 65  | e     |
| 6       | 6   | [ACKNOWLEDGE]          | 38      | 26  | &       | 70      | 46  | F    | 102     | 66  | f     |
| 7       | 7   | [BELL]                 | 39      | 27  | '       | 71      | 47  | G    | 103     | 67  | g     |
| 8       | 8   | [BACKSPACE]            | 40      | 28  | (       | 72      | 48  | H    | 104     | 68  | h     |
| 9       | 9   | [HORIZONTAL TAB]       | 41      | 29  | )       | 73      | 49  | I    | 105     | 69  | i     |
| 10      | A   | [LINE FEED]            | 42      | 2A  | *       | 74      | 4A  | J    | 106     | 6A  | j     |
| 11      | B   | [VERTICAL TAB]         | 43      | 2B  | +       | 75      | 4B  | K    | 107     | 6B  | k     |
| 12      | C   | [FORM FEED]            | 44      | 2C  | ,       | 76      | 4C  | L    | 108     | 6C  | l     |
| 13      | D   | [CARRIAGE RETURN]      | 45      | 2D  | -       | 77      | 4D  | M    | 109     | 6D  | m     |
| 14      | E   | [SHIFT OUT]            | 46      | 2E  | .       | 78      | 4E  | N    | 110     | 6E  | n     |
| 15      | F   | [SHIFT IN]             | 47      | 2F  | /       | 79      | 4F  | O    | 111     | 6F  | o     |
| 16      | 10  | [DATA LINK ESCAPE]     | 48      | 30  | 0       | 80      | 50  | P    | 112     | 70  | p     |
| 17      | 11  | [DEVICE CONTROL 1]     | 49      | 31  | 1       | 81      | 51  | Q    | 113     | 71  | q     |
| 18      | 12  | [DEVICE CONTROL 2]     | 50      | 32  | 2       | 82      | 52  | R    | 114     | 72  | r     |
| 19      | 13  | [DEVICE CONTROL 3]     | 51      | 33  | 3       | 83      | 53  | S    | 115     | 73  | s     |
| 20      | 14  | [DEVICE CONTROL 4]     | 52      | 34  | 4       | 84      | 54  | T    | 116     | 74  | t     |
| 21      | 15  | [NEGATIVE ACKNOWLEDGE] | 53      | 35  | 5       | 85      | 55  | U    | 117     | 75  | u     |
| 22      | 16  | [SYNCHRONOUS IDLE]     | 54      | 36  | 6       | 86      | 56  | V    | 118     | 76  | v     |
| 23      | 17  | [ENG OF TRANS. BLOCK]  | 55      | 37  | 7       | 87      | 57  | W    | 119     | 77  | w     |
| 24      | 18  | [CANCEL]               | 56      | 38  | 8       | 88      | 58  | X    | 120     | 78  | x     |
| 25      | 19  | [END OF MEDIUM]        | 57      | 39  | 9       | 89      | 59  | Y    | 121     | 79  | y     |
| 26      | 1A  | [SUBSTITUTE]           | 58      | 3A  | :       | 90      | 5A  | Z    | 122     | 7A  | z     |
| 27      | 1B  | [ESCAPE]               | 59      | 3B  | ;       | 91      | 5B  | [    | 123     | 7B  | {     |
| 28      | 1C  | [FILE SEPARATOR]       | 60      | 3C  | <       | 92      | 5C  | \    | 124     | 7C  |       |
| 29      | 1D  | [GROUP SEPARATOR]      | 61      | 3D  | =       | 93      | 5D  | ]    | 125     | 7D  | }     |
| 30      | 1E  | [RECORD SEPARATOR]     | 62      | 3E  | >       | 94      | 5E  | ^    | 126     | 7E  | ~     |
| 31      | 1F  | [UNIT SEPARATOR]       | 63      | 3F  | ?       | 95      | 5F  | _    | 127     | 7F  | [DEL] |

Figure 1: ASCII Tabelle

## The Block Cipher DES

DES is based on the application of substitution boxes and a structure called the Feistel network. For secure encryption, the S-boxes must meet the following requirements to ensure a secure cipher:

**Completeness** Each bit in the output depends on every bit in the input.

**Avalanche Effect** Changing one bit in the input changes, on average, half of the output bits.

**Non-linearity** No output bit is linearly dependent on a single input bit.

**Correlation Immunity** As long as only a part of the input bits is known, no inference can be made about the output bits, and vice versa.

Table 2 shows S-Box #5 of DES. The DES S-boxes were constructed to satisfy the criteria listed above.

## SBox 5 (DES)

| $S_5$       |    | Mittlere 4 Bits des Eingabewertes |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|-------------|----|-----------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
|             |    | 0000                              | 0001 | 0010 | 0011 | 0100 | 0101 | 0110 | 0111 | 1000 | 1001 | 1010 | 1011 | 1100 | 1101 | 1110 | 1111 |
| Äußere Bits | 00 | 0010                              | 1100 | 0100 | 0001 | 0111 | 1010 | 1011 | 0110 | 1000 | 0101 | 0011 | 1111 | 1101 | 0000 | 1110 | 1001 |
|             | 01 | 1110                              | 1011 | 0010 | 1100 | 0100 | 0111 | 1101 | 0001 | 0101 | 0000 | 1111 | 1010 | 0011 | 1001 | 1000 | 0110 |
|             | 10 | 0100                              | 0010 | 0001 | 1011 | 1010 | 1101 | 0111 | 1000 | 1111 | 1001 | 1100 | 0101 | 0110 | 0011 | 0000 | 1110 |
|             | 11 | 1011                              | 1000 | 1100 | 0111 | 0001 | 1110 | 0010 | 1101 | 0110 | 1111 | 0000 | 1001 | 1010 | 0100 | 0101 | 0011 |

Figure 2: S-Box #5 of DES. A 6-bit input value is taken, which produces a 4-bit output value through substitution. Example: **011011** has outer bits **01** and inner bits **1101**, resulting in the substitution value 1001 according to the table (highlighted in yellow).

## Questions on DES

### Exercise 3.

1. What key length was originally proposed for DES, and what (effective) key length was ultimately chosen? Who was responsible for this change?
2. What do the terms confusion and diffusion mean, and which components of DES implement these functions?
3. Draw a diagram of a Feistel network.

## Non-linearity of S-Boxes

| $S_1$ | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
|-------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0     | 14 | 04 | 13 | 01 | 02 | 15 | 11 | 08 | 03 | 10 | 06 | 12 | 05 | 09 | 00 | 07 |
| 1     | 00 | 15 | 07 | 04 | 14 | 02 | 13 | 01 | 10 | 06 | 12 | 11 | 09 | 05 | 03 | 08 |
| 2     | 04 | 01 | 14 | 08 | 13 | 06 | 02 | 11 | 15 | 12 | 09 | 07 | 03 | 10 | 05 | 00 |
| 3     | 15 | 12 | 08 | 02 | 04 | 09 | 01 | 07 | 05 | 11 | 03 | 14 | 10 | 00 | 06 | 13 |

Table 2: S-Box #1 of DES. The values in the table are in decimal notation!

Use DES S-Box #1 for the following tasks. A copy of the S-box can be found in Table 2.

An S-box  $S_i$  behaves linearly if the following holds:  
 $S_i(x_1) \oplus S_i(x_2) = S_i(x_1 \oplus x_2)$   
 where the operator  $\oplus$  denotes bitwise XOR.

### Exercise 4.

1. One important property of DES is the non-linearity of its S-boxes. Show, for the following input bits, that the S-box is not linear:
  - $x_1 = 000000$  and  $x_2 = 000001$
  - $x_1 = 111111$  and  $x_2 = 100000$
  - $x_1 = 101010$  and  $x_2 = 010101$

## Bonus Exercise: Weak Keys

This task requires time and a good understanding of DES. Therefore, we will discuss it at the beginning of the next exercise session.

A DES key is considered weak if encryption and decryption are identical operations:

A DES key is weak if the following condition is fulfilled:

$$DES_{K_w}(x) = (DES_{K_w}(x))^{-1}, \text{ for all } x.$$

**Hint:** Look closely at the subkeys, especially the round keys of DES. Draw the DES structure for at least 2 rounds (see Question 3).

### Exercise 5.

1. Describe what values the subkeys  $K_i$  would need to take over the 16 DES rounds for the above weak key equation to hold.
2. There are 4 weak DES keys. Which are they?
3. What is the probability that a randomly chosen DES key is a weak key?